Name:		Centre/Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H1 BIOLOGY

8876/02

Paper 2 Structured and Free-response Questions

13 September 2021

2 hours

Additional Material: Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write your centre number, index number, name and class at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the question paper.

Section B
Answer any **one** question in the spaces provided in the answer booklet.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**21**
2	**9**
3	**10**
4	**5**
Section B	
5 / 6	**15**
Total	**60**

This document consists of **14** printed pages (including this cover page) and **2** blank pages.

For Examiner's use

Section A: Structured Questions (45 marks)
Answer **all** questions in this section.

Question 1

Fig 1.1 shows the structure of a mammalian eye. A section of the retina is boxed up for a close up look.

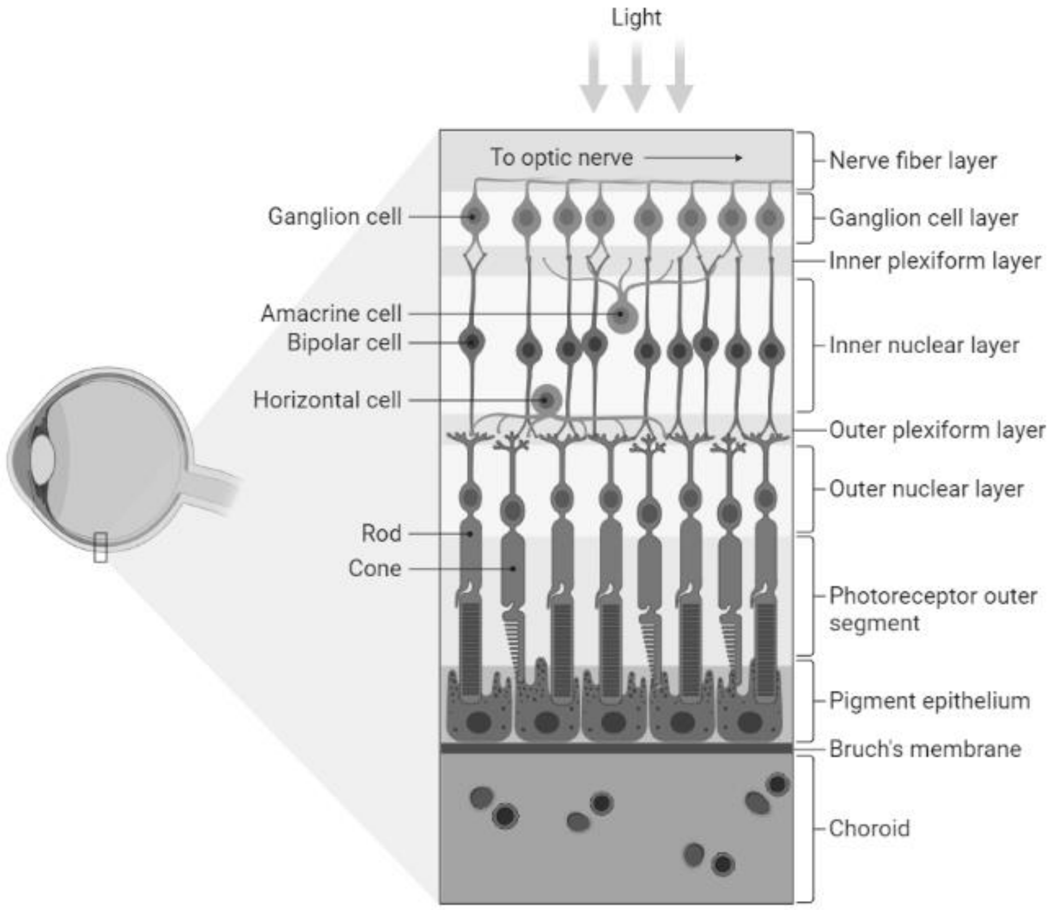

Fig 1.1

(a) Explain how Fig 1.1 supports the cell theory.

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....

....

.... [2]

Fig 1.2 shows a microscopic image of part of a rod cell, one of the photoreceptor cells in the retina.

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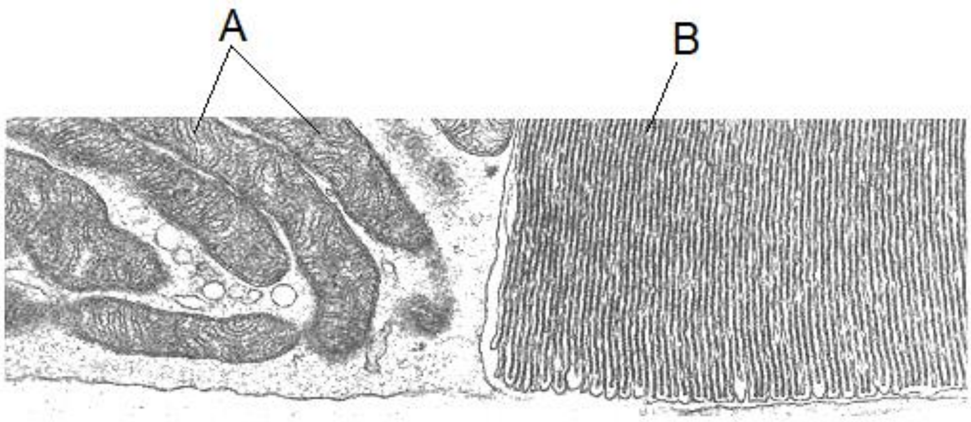

Fig 1.2

(b) **(i)** With reference to Fig 1.2, identify organelle **A**.

.... [1]

(ii) Using your knowledge, explain how organelle **A** is suited for its function in a cell.

....

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....

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....

.... [3]

(c) Fig 1.3 shows a drawing of a rod cell and an organelle found in plant cells. Structure **B** can also be seen in Fig 1.2.

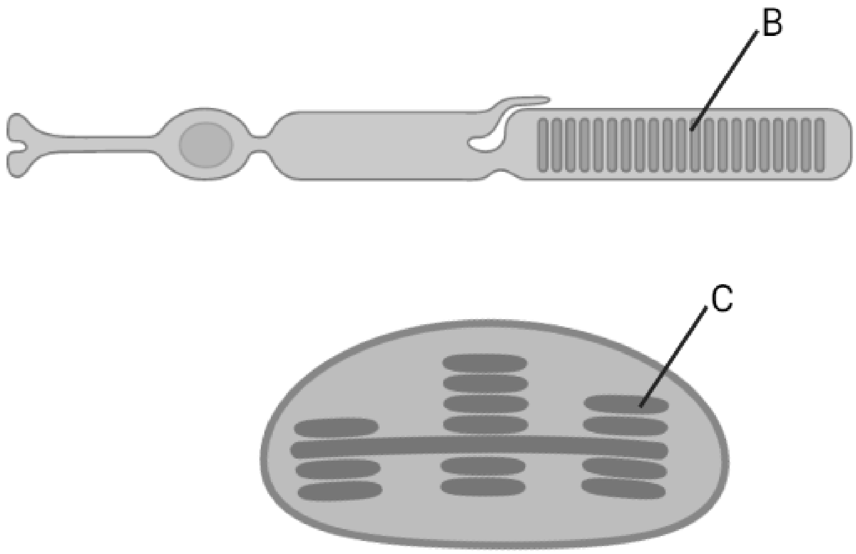

Fig 1.3

Structure **B** is a disc found in the outer segment of the rod cell. The disc is membrane-enclosed sac packed with photo-sensitive molecules known as *rhodopsin*.

Identify structure **C** and compare structures **B** and **C**.

....

....

....

....

....

....

.... [3]

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Fig 1.4 shows a section of structure **C** membrane.

For Examiner's use

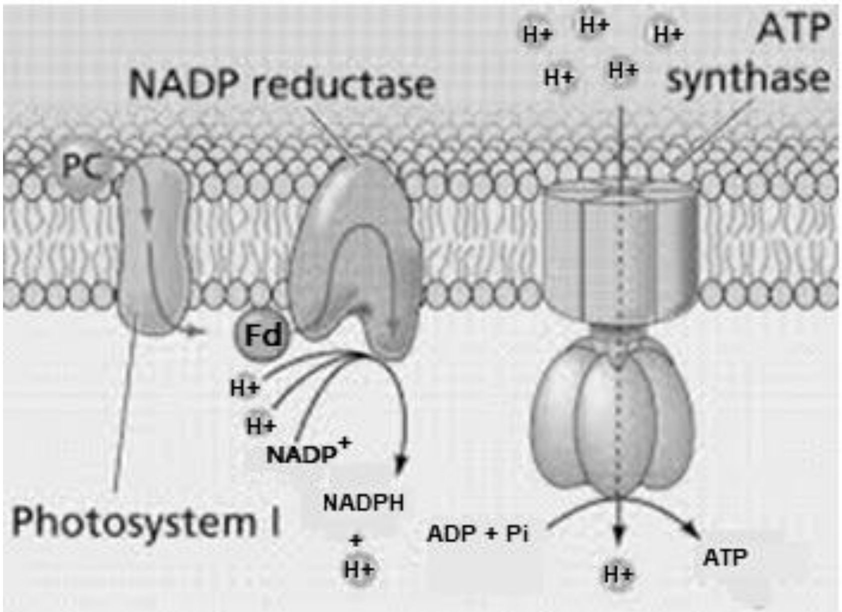

Fig 1.4

(d) With reference to Fig 1.4,

(i) explain how a visible feature of the membrane supports the fluid mosaic model.

....

....

....

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....

....

.... [3]

For Examiner's use

(ii) explain how NADP reductase is stabilised in structure **C** membrane.

[2]

(iii) briefly explain the mode of action of the enzyme NADP reductase.

[3]

(iv) explain how an increase in temperature increases the rate of photosynthesis.

[4]

[Total: 21]

For Examiner's use

Question 2

Fig 2 depicts a model of cancer development.

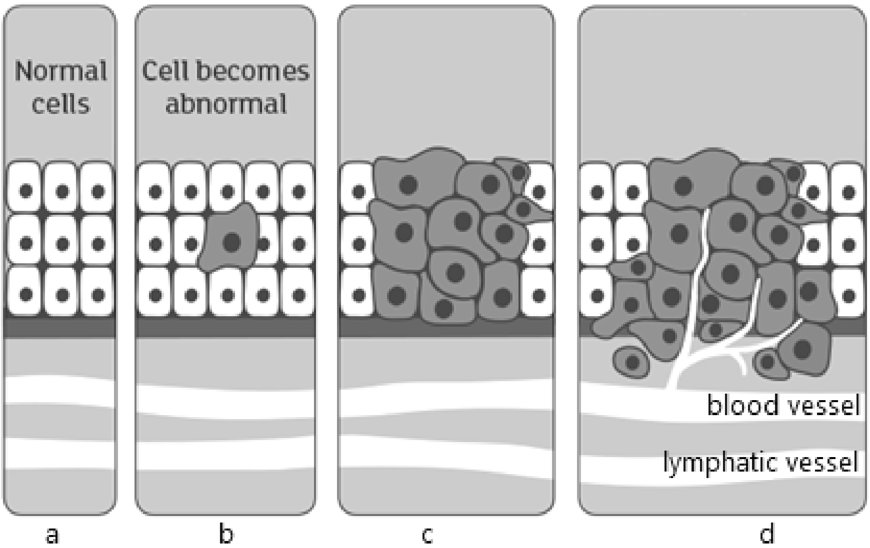

Fig 2

(a) With reference to Fig 2, explain the development of cancer.

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....

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....

....

.... [4]

For Examiner's use

Recently, scientists discovered that some cells within tumors replace glucose with lipids as an energy source in order to multiply.

(b) **(i)** Lipids require more oxygen than glucose to be completely oxidized. Explain why.

....

.... [1]

(ii) Explain why it is better for tumor cells to oxidise lipids than glucose.

....

....

....

.... [2]

In another study investigating the differences between benign and malignant tumor cells, it was discovered that malignant tumor cells have cell surface membrane that is more fluid than benign cells.

(c) **(i)** State one possible structural difference between the cell surface membrane of malignant and benign tumor cells.

.... [1]

(ii) Suggest an advantage of higher membrane fluidity in the transformation from benign to malignant tumor cells.

....

.... [1]

[Total: 9]

For Examiner's use

Question 3

In rabbits, there are several alleles of a gene that codes for an enzyme in the synthesis of pigments. The different alleles of this gene result in different fur colour. The alleles are listed below in order of dominance, with **C** being the most dominant.

C = full colour
C^{ch} = chinchilla
C^{h} = Himalayan
C^{a} = albino

(a) Explain how mutation to the C gene gives rise to the C^{a} allele coding for the albino phenotype.

....

....

....

....

.... [2]

(b) Explain how recessive alleles are preserved in a natural population.

....

....

....

.... [2]

The results of a test cross on a full coloured male rabbit revealed that 50% of the offspring had chinchilla fur. This male rabbit was crossed with a female heterozygote Himalayan rabbit.

For Examiner's use

(c) **(i)** Using a genetic diagram, show the result of this cross.

[3]

(ii) State the probability of obtaining a Himalayan offspring from this cross.

.... [1]

For Examiner's use

The phenotype of an individual is determined by its genotype. However, the environment may influence gene expression.

(d) Explain how temperature affects their fur colour.

[2]

[Total: 10]

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For Examiner's use

Question 4

Evolutionary changes in the beaks of Darwin's finches have been instrumental in the adaptive radiation of these birds in the Galapagos Islands.

There are a dozen species that are closely related to one another showing remarkable differences in beak shape and size, and in eating habits. They have all arisen from a common ancestor, but have evolved to exploit different food sources.

The precise dimensions (length, depth and width) of each species' beak are crucial to their survival, and fluctuations in the environment lead to selection that changes the relative success of birds with various beak shapes.

The molecular basis for variation in beak size and shape is due to two genes:

the gene for **bone morphogenetic protein 4 (BMP4)** that encodes a protein which allows for cell-to-cell communication that is expressed more during the embryonic development of the deep and wide beaks of ground finches than during the narrower beaks.

the gene for **calmodulin (CaM)** that encodes a calcium signalling protein that is expressed at much higher levels in the long and pointed beaks of cactus finch embryos.

Fig 4.1 shows how these two genes interact to produce variation in beak size and shape in the different finch species.

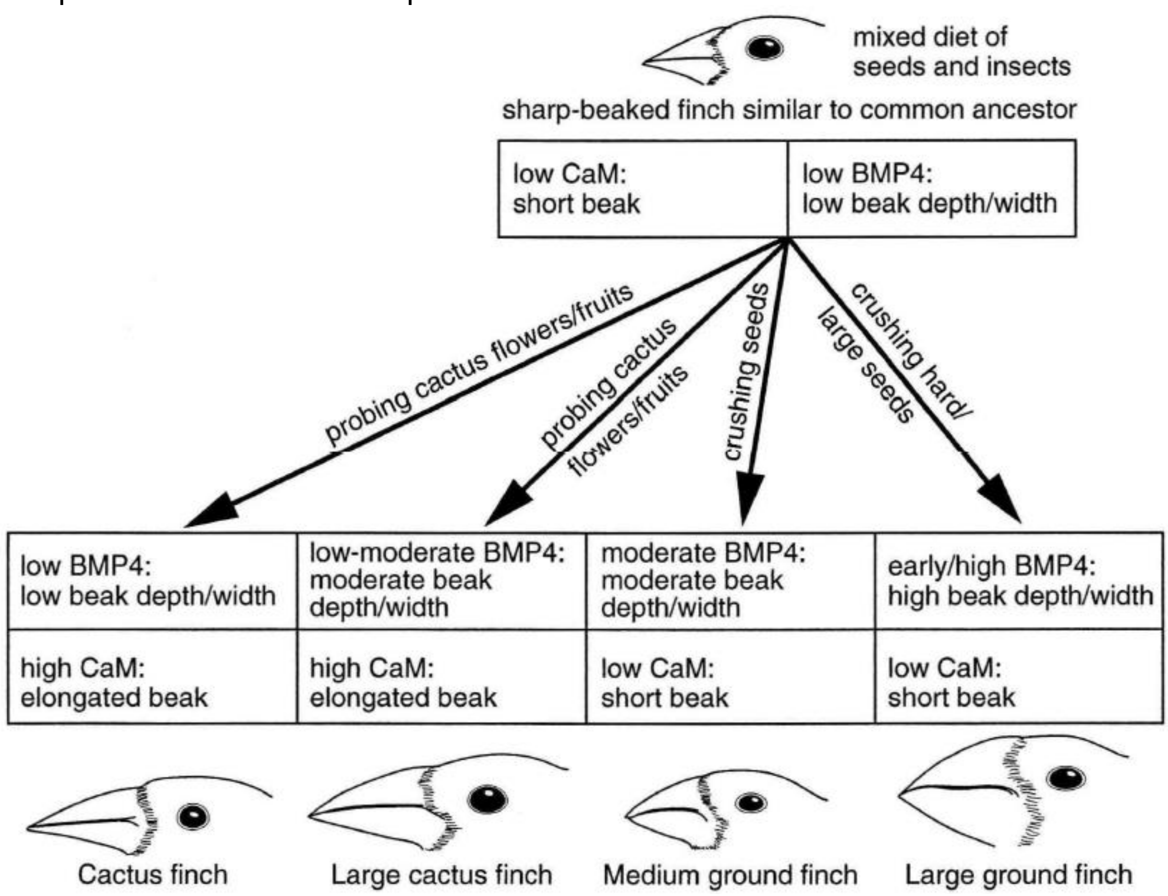

Fig 4.1

Explain how the difference in beak size and shape between species may have arisen.

For Examiner's use

[5]

[Total: 5]

Section B: Free-response Questions (15 marks)

Answer **ONE** question.
Write your answers on the separate answer booklet provided.
Your answers should be illustrated by large, clearly labelled diagrams, wherever appropriate.

Your answer must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.
A **NIL RETURN** is required.

Question 5

(a) Discuss how DNA changes during gamete formation. [10]

(b) Compare structure and roles of mRNA and tRNA. [5]

Total: [15]

OR

Question 6

(a) Discuss how changes in protein conformation are important. [10]

(b) Compare replication and transcription. [5]

Total: [15]

DUNMAN HIGH SCHOOL
DHP YR 6
H2 BIOLOGY

Year 6 H1 Preliminary Exam 2021
Suggested Answers

Paper 2 Section B

Question 1			
(a)	The retina layer is composed of multiple cell types + name at least 2 cell types; Supporting the cell theory that living organisms are made composed of cells;		[2]
(b)	(i)	Mitochondria;	[1]
	(ii)	Highly folded inner membrane / cristae; Enables many electron carriers / electron transport chains / ATP synthase to be embedded; For high rate of ATP synthesis; For cellular activities; Max 3	[3]
(c)		Thylakoid; Any 2 below - Both are single membrane bound; Both contains many photo-sensitive molecules; B is located within the cytoplasm of the cell while C is located within a chloroplast in a cell; B is found in animal cells while C is found in plant cells;	[3]
(d)	(i)	Random arrangement of proteins embedded in membrane; Proteins and lipids able to move rapidly and laterally in the plane of the membrane; ref to NADP reductase and ATP synthase; OR Ref to PC able to move towards photosystem I;	[3]
	(ii)	Globular structure with hydrophobic regions interacting with hydrophobic core of the phospholipid bilayer;	[2]

		Hydrophilic regions facing the aqueous thylakoid space and stroma; OR Hydrophilic regions interacting with polar phosphate head of phospholipid;	
	(iii)	<u>Active site</u> specific / complementary; to any one substrate - Fd / electron, H^+ or $NADP^+$; Form enzyme-substrate complex + reduces the <u>activation energy</u>; Catalyses the reduction of $NADP^+$ with the addition of H^+ and electron; Any 3	[3]
	(iv)	Increase kinetic energy of enzyme and substrate; Ref to one enzyme and respective substrate; Eg NADP reductase and $NADP^+$ / electron / H^+ / Fd OR ATP synthase and ADP / Pi Increase rate of effective collision; Increase in rate of enzyme-substrate complex formation;	[4]

Question 2			
(a)	Transformation of normal cell to abnormal cell due to loss of checkpoint control / mutations in checkpoint control genes; Abnormal cell divides uncontrollably + forming a tumor; ref to accumulation of mutations; Ref to loss of contact inhibition; Ref to angiogenesis; Ref to loss of anchorage; Ref to metastasis; Max 4		[4]
(b)	(i)	Lipids have more C-H bonds than glucose;	[1]
	(ii)	Oxidation of lipids produce two times more energy per gram; To support the high rate of cell division which requires energy;	[2]
(c)	(i)	More cholesterol; OR More unsaturated fatty acids;	[1]
	(ii)	Enable the cell to squeeze endothelial layer into and out of blood vessel /AVP;	[1]

Question 3

(a)	Mutation results in a change in amino acid sequence and conformation of the enzyme coded for; Hence enzyme is non-function and unable to synthesize pigment;	[2]
(b)	Individuals are <u>diploid</u> and genetic variation can be maintained in the form of recessive alleles in heterozygotes, allowing recessive alleles to persist in the population.; There may be balancing selection / frequency-dependent selection such that the individuals that are too common declines as they are targeted by predators, preserving polymorphism.; There is variation in environment / habitats, allowing individuals with different genotypes and phenotypes to survive and reproduce, preserving polymorphism.; Due to <u>heterozygote advantage</u>, recessive alleles can be preserved in heterozygotes.; There is neutral selection / genetic variation that confer no selective advantage, maintaining the frequency of recessive alleles in population.; <u>Max 2</u>	[2]
(c)	(i)	[3]

Parental phenotypes:	Full colour male	X	Himalayan colour female	
Parental genotype:	CC^{ch}	X	C^hC^a	
Gametes formed:	C C^{ch}	X	C^h	C^a

F₁ genotypes:

	C	C^{ch}
C^h	CC^h	$C^{ch}C^h$
C^a	CC^a	$C^{ch}C^a$

F_1 phenotypes

C_ :	C^{ch}_
full color :	Chinchilla
1 :	1

1m – parental phenotype + genotypes;
1m – gametes (circled);
1m – F1 genotypes and phenotypes;

	(ii) 0;	[1]
(d)	Heat from environment prevents the development of black fur.; Tyrosinase required for melanin production is heat sensitive and is denatured at higher temperatures. Hence, fur-producing cells do not produce melanin and fur appears white.;	[2]

Question 4	
There is variation in beak shape and size in the population of common ancestor / sharp-beaked finch; The variation is due to different expression levels in *CaM* and *BMP4* genes + as a result of random gene mutations and sexual reproduction; Different food sources / availability of food acts as a selection pressure for depth, width and length of beaks of finches.; Ref to any example of food source and beak shape selected; These finches were able to survive and reproduce, passing down alleles for desried traits to their offspring, increasing in frequency over time.; Over a long period of time, the accumulation of multiple changes in genetic makeup (including *BAMP4* and *CaM*) results in the formation of cactus finch, a new species.; Max 5	[5]

<u>Paper 2 Section C</u>

5 a Discuss how DNA changes during gamete formation. [10]

1. DNA **doubles via DNA <u>replication</u>** during **S phase** / interphase;

2. DNA **sequence may be mutated due to mistakes** during replication;

3. During **prophase I**, chromatin structure of DNA **condenses** into chromosomes;

4. Now, the chromosome is made up of **2** genetically identical **sister chromatids held together at the centromere**;

5. Homologous chromosomes **pair up** during **prophase I** via synapsis, forming a bivalent / tetrad;

6. **Chiasmata formation** between the homologous chromosomes allow **crossing over** to take place;

7. This results in **genetic recombination**;

8. Whereby the chromosomes comprise of **chromatids that are no longer genetically identical**;

9. During **anaphase I**, the **pairs of homologous chromosomes are separated** as they are pulled to opposite poles with the shortening of spindle fibres;

10. The **number of chromosomes in each nuclei is now half** of the parental number of chromosomes (n);

11. At **anaphase II**, the **non-identical sister chromatids are separated**;

12. as **centromere divides**;

13. **Structural aberrations** to the chromosomes might arise during meiosis;
14. Eg. When sister chromatids are pulled apart during anaphase II, a piece of the chromatid might break apart, leading to deletions / AVP ;

15. At **telophase II**, the chromosomes **de-condense** back to the thread-like chromatin structure;

Max 9

5 b Compare structure and roles of mRNA and tRNA. [5]

<u>Similarities</u>

1. [Structure] Both are **single stranded**;
2. [Role] **Both binds to ribosomes** during translation / AW;

R! Both comprise of ribonucleotides;

<u>Differences</u>

	mRNA	tRNA
3.	[Structure] linear	Folds back upon itself to form hairpin loops / double stranded regions;
4.	[Structure] 3' polyA tail	3' CCA end;
5.	[Structure] 5' 7-methylguanosine cap	Absent;
6.	[Structure] Carries codons to be recognized by tRNA during translation OR Carries codons to specify sequence of amino acids in a polypeptide chain	Carries amino acids to the ribosome during translation;

Max 4

6 a Discuss how changes in protein conformation are important. [10]

Structural changes in proteins that have important role in normal function:

Structural change	Importance
Transport across membrane	
1. Alternating conformation of protein carriers in membranes	2. Enables facilitated diffusion of large polar molecules / ions down their concentration gradient
3. Opening and closing of gated channel proteins in membranes	4. Allows control of movement of substances into and out of cells 5. E.g. gated calcium channels in ER to control release of calcium
6. Protein pumps in membrane use hydrolysis of ATP for change in conformation	7. Enables active transport of large polar molecules / ions against their concentration gradient
8. ATP synthase changes conformation as protons diffuse through it	9. Allowing the synthesis of ATP
Transport of oxygen	
10. Binding of oxygen to one subunit of Hb results in change in conformation in the remaining 3 subunits	11. Increases the affinity of the remaining subunits for binding oxygen;
Enzymes	
12. Induced fit – binding of substrate to enzyme active site results in a change in conformation of the enzyme to fit the substrate more tightly	13. Applies strain to bonds in substrate / brings catalytic residues in position / AW to lower activation energy for reaction
Mitosis and Meiosis	
14. Lengthening or shortening of spindle fibres	15. Shortening of spindle fibres attached to centromeres allows equal separation of chromosomes / chromatids into daughter cells 16. Lengthening allows elongation of cell prior to divison
Cytoskeleton	
17. Polymerization of cytoskeleton / microtubules / microfilament / ORA;	18. AVP; Eg. For formation of pseudopodia during phagocytosis

6 b Compare DNA replication and transcription. [5]

Similarities

1. Both use a template to synthesise complementary strand
   OR
   complementary base pairing

2. Synthesis direction in 5’ to 3’ direction
   OR
   template read in 3’ to 5’ direction

3. Phosphodiester bonds formed

	DNA replication	**transcription**
4. template	Both strands of DNA used as template	Only 1 strand of the double stranded DNA used as template
5. Length of template copied	**Entire DNA molecule** replicated	Only **gene** is transcribed
6. Enzyme	DNA polymerase	RNA polymerase
7. Molecule synthesised	DNA	RNA
8. Monomers used	deoxyribonucleotides	Ribonucleotides
9. Phase of cell cycle	Occurs in S phase	Occurs in G0, G1 and G2 phase